

FOST TRAINING RULE OF THE ROAD TEST JUN 16

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(G) Vol I and the Mariners Handbook (NP100).

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the OOW training records for inspection by FOST navigation staff. The exam should take no longer than 1 hr.

In this test, any change in the status of the ship or of the environment will be indicated in the script. Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90%.

You are the OOW on RFA MOUNTS BAY (length 177 metres), alongside Gibraltar preparing to sail for SOLAS specific internal training. Weather is fair and visibility is good. Wind is W at 15kts and is forecast to reduce with a risk of Fog overnight. There are various serials planned throughout the day making it seem like an OST period.

The ship closes up SSD and the un-berthing proceeds without incident; the CO hands you the ship and you commence the outbound pilotage.

1. On taking the ship you sight a small power-driven vessel at green 45 at 6 cables crossing the channel. Her bearing is steady. Assuming it is a Narrow channel and that she is already impeding your safe passage:

- a. Who is the give-way vessel and why (R8fiii, R15) **(2)**

- b. What action, if any, are you required to take (R15, R34D) **(1)**

- c. What is the obligation of the other vessel given she had already impeded your passage? (R.8fiii) **(2)**

The above situation is resolved and the pilotage continues.

2. The OOW reports an outbound merchant vessel closing slowly at red 170 at 2nm.
- a. What situation is this and why? (R. 13b) **(2)**

b. If you alter course 70° to port in accordance with your navigation track how would this change the situation? (R. 13d) **(2)**

c. Your CO, keen to proceed into open water, suggests an increase in speed to 18 knots. What do the rules say about safe speed? List the pertinent considerations before giving the conning order? (R 6a, 6b) **(7) State rule and list 6 reasons**

3. You clear pilotage waters and fall out SSD. At red 20 at 5nm you sight two small power driven vessels. Their range is closing and both of their bearings are drawing very slowly right. They physically appear to be fishing vessels underway and you confirm this via their AIS transmissions. As you get closer you sight two cones apexes together on both vessels.

a. What do these shapes mean? (R. 26b) **(1)**

b. How is this category of vessel defined? (R. 3d) **(1)**

c. What lights would these vessels show by night? (R. 26b) **(3)**

d. What sound signals would this category of vessel make in restricted visibility? (R. 35c) **(2)**

e. In accordance with Rule 18 you make a bold alteration of course to starboard to keep clear of the fishing vessels. Should you use sound signals prior to this manoeuvre? Justify your answer. (R. 34a) **(3)**

You clear these vessels and continue to proceed along the course of your nav track. You are proceeding to an RV to conduct a PASSEX with SNMG1 TG. Unfortunately you see a Fog bank ahead and visibility begins to reduce to 1nm.

4. You detect a vessel by radar alone at red 20 at 1.5nm.

a. List your reactions to this contact. (R. 19d) **(3)**

b. You assess the contact above to be drawing rapidly left with a CPA in excess of 2nm. You now hold a small power driven vessel visually at green 40 at 9c; you are 60° on her port bow and her bearing is drawing very slowly left. Does Rule 19 still apply? Explain how you would manoeuvre for this contact. (R. 19a, 15) **(3)**

c. You make a shipping report to the CO and take the correct action in good time. Continuing along your nav track, you hear a fog signal to port consisting of one prolonged blast followed by two short blasts. What type of vessel could this be? (R. 35c) **(3-half mark each)**

d. You check the radar and hold no steady contacts on the port side other than a faint echo at R20, range 5 cables. You cannot track the echo effectively or generate an accurate CPA using the radar. What action should you take in accordance with the rules? (R. 19e) **(2)**

The contact closes to 1 mile and you sight the following lights:

R
G

R

- e. What type of vessel is this? (R. 25a/c) **(1)**

5. The vessel passes clear down your port side and you notice that visibility is rapidly improving as you exit the fog bank. You assess visibility as 10nm and inform the CO who is happy for you now to increase speed to meet the required SOA. As you come up in speed you sight a small cargo vessel at G30, range 3nm. You assess that you are 40 degrees on her port bow and her bearing is steady.

- a. In accordance with the Rules what action should you take? **(R. 15)** (2)

- b. What should the cargo vessel do? **(R. 17)** (1)

- c. The vessel is now at 5 cables and you have not taken any action to avoid collision. You decide, given the closing range, that urgent action is required to avoid collision and inform the CO. What could you decide to do? **(R. 8e)** (1)

- d. How should any action be taken to avoid collision in accordance with the rules? **(R. 8a)** (1)

6. The cargo vessel takes avoiding action in accordance with Rule 17b and makes a bold alteration of course to starboard. A collision narrowly avoided, the vessel passes 3 cables clear down your stbd side. The CO comes to the bridge to debrief you on the recent incident. He stresses the importance of the Rules and the requirement to act early to generate 'thinking time' in a developing situation. You both reflect on the shipping situation and agree that a simple alteration of course would have resolved the problem early. If taking action to avoid collision by alteration of course alone, what should you consider? **(R. 8c)** (2)

The rest of the watch passes without incident.

7. You approach the formation of SNMG1 units and note all other ships are in station.

a. What does ATP1G state about how ships not in station should manoeuvre? (ATP1G) **(1)**

b. What should ships in station consider in such a situation? (ATP1G) **(1)**

c. Once in station yourself you note, the Italian aircraft carrier CAVOUR has commenced fixed wing operations. What is the order of priority applicable to Allied naval ships? (ATP1G) **(6)**

d. Do the above priorities take precedence over the International Regulations for Preventing Collisions at Sea? (ATP1G) **(2)**

The PASSEX completes at sunset. As you break away you note that the ITS CAVOUR does not have navigation lights in standard positions.

8. What do the Rules say about the operation of special rules with respect to additional station or signal lights? (R. 1c) **(2)**

You are preparing night orders after a long day on the bridge when the CAVOUR sends Flag 5 with a time and you hear her sounding 6 short blasts.

9. What does this mean? (ATP 1G) **(1)**

You report this to the CO and head down below for dinner.

You are tasked to proceed alongside to collect another batch of SOLAS stores prior to heading east. After a quiet evening you are up on the bridge early to make final preparations for SSD.

10. You sight a vessel in the early morning twilight proceeding towards the harbour showing white over red masthead lights in addition to sidelights and sternlight.

- a. What can you deduce about this vessel? (R. 29a) **(2)**

- b. What identity signal would this vessel sound in restricted visibility? (R. 35k) **(2)**

Total / 62

ANSWERS

You are the OOW on RFA MOUNTS BAY (length 177 metres), alongside Gibraltar preparing to sail for SOLAS specific internal training. Weather is fair and visibility is good. Wind is W at 15kts and is forecast to reduce with a risk of Fog overnight. There are various serials planned throughout the day making it seem like an OST period.

The ship closes up SSD and the un-berthing proceeds without incident; the CO hands you the ship and you commence the outbound pilotage.

1. On taking the ship you sight a small power-driven vessel at green 45 at 6 cables crossing the channel. Her bearing is steady. Assuming it is a Narrow channel and that she is already impeding your safe passage:
 - a. Who is the give-way vessel and why (R8fiii, R15) **(2)**
 - b. What action, if any, are you required to take (R15, R34D) **(1)**
 - c. What is the obligation of the other vessel given she had already impeded your passage? (R.8fiii) **(2)**

a. She should not impede your passage but at a range of 6c, has already done so. A vessel the passage of which is not to be impeded remains fully obliged to comply with the Rules of this part when two vessels are approaching so as to involve risk of collision (1). You are therefore still required to give way (1).

b. Since you are in a narrow channel, the best course of action is to reduce speed (1).

c. A vessel required not to impede the passage or safe passage of another vessel is not relieved of this obligation if approaching the other vessel so as to involve risk of collision and shall, when taking action, have full regard to the action which may be required by the Rules of this part (2).

The above situation is resolved and the pilotage continues.

2. The OOW reports an outbound merchant vessel closing slowly at red 170 at 2nm.
 - a. What situation is this and why? (R. 13b) **(2)**

Overtaking situation (1), the merchant vessel is coming up with your vessel from a direction more than 22.5° abaft your beam (1).

b. If you alter course 70° to port in accordance with your navigation track how would this change the situation? (R. 13d) (2)

It wouldn't change any obligation under Rule 13 (1), any subsequent alteration of bearing between the two vessels shall not make the overtaking vessel a crossing vessel within the meaning of these rules or relieve her of the duty of keeping clear of the overtaken vessel until she is finally past and clear (1).

c. Your CO, keen to proceed into open water, suggests an increase in speed to 18 knots. What do the rules say about safe speed? (1) and list 6 considerations before giving the conning order? (R 6a, 6b) (6)

Every vessel shall at all times proceed at a safe speed so that she can take proper and effective action to avoid collision and be stopped within a distance appropriate to the prevailing circumstances and conditions (1).

6 of the following (1 mark each):

(i) the state of visibility;

(ii) the traffic density including concentrations of fishing vessels or any other vessels;

(iii) the manoeuvrability of the vessel with special reference to stopping distance and turning ability in the prevailing conditions;

(v) the state of wind, sea and current, and the proximity of navigational hazards;

(vi) the draught in relation to the available depth of water.

(i) the characteristics, efficiency and limitations of the radar equipment;

(ii) any constraints imposed by the radar range scale in use;

(iii) the effect on radar detection of the sea state, weather and other sources of interference;

(iv) the possibility that small vessels, ice and other floating objects may not be detected by radar at an adequate range;

(v) the number, location and movement of vessels detected by radar;

(vi) the more exact assessment of the visibility that may be possible when radar is used to determine the range of vessels or other objects in the vicinity.

3. You clear pilotage waters and fall out SSD. At red 20 at 5nm you sight two small power driven vessels. Their range is closing and both of their bearings are drawing very slowly right. They physically appear to be fishing vessels underway and you confirm this via their AIS transmissions. As you get closer you sight two cones apexes together on both vessels.

a. What do these shapes mean? (R. 26b) **(1)**

A vessel engaged in fishing.

b. How is this category of vessel defined? (R. 3d) **(1)**

The term "vessel engaged in fishing" means any vessel fishing with nets, lines, trawls or other fishing apparatus which restrict manoeuvrability, but does not include a vessel fishing with trolling lines or other fishing apparatus which do not restrict manoeuvrability.

Definition of trawling also accepted.

c. What lights would these vessels show by night? (R. 26b) **(3)**

A vessel engaged in fishing, other than trawling, shall exhibit:

(i) two all-round lights in a vertical line, the upper being red and the lower white, or a shape consisting of two cones with apexes together in a vertical line one above the other (1);

(ii) when there is outlying gear extending more than 150 metres horizontally from the vessel, an all-round white light or a cone apex upwards in the direction of the gear;

(iii) when making way through the water, in addition to the lights prescribed in this paragraph, sidelights and a sternlight (2).

Lights for trawling also accepted.

d. What sound signals would this category of vessel make in restricted visibility? (R. 35c) (2)

One prolonged followed by two short blasts (1) at intervals of not more than 2 minutes(1).

e. In accordance with Rule 18 you make a bold alteration of course to starboard to keep clear of the fishing vessels. Should you use sound signals prior to this manoeuvre? Justify your answer. (R. 34a) (3)

Yes (1), when vessels are in sight of one another, a power-driven vessel underway, when manoeuvring as authorised or required by these Rules, shall indicate that manoeuvre by signals on her whistle (2).

You clear these vessels and continue to proceed along the course of your nav track. You are proceeding to an RV to conduct a PASSEX with SNMG1 TG. Unfortunately you see a Fog bank ahead and visibility begins to reduce to 1nm.

4. You detect a vessel by radar alone at red 20 at 1.5nm.

b. List your reactions to this contact. (R. 19d) (3)

A vessel which detects by radar alone the presence of another vessel shall determine if a close-quarters situation is developing and/or risk of collision exists (1). If so, she shall take avoiding action in ample time, provided that when such action consists of an alteration in course, so far as possible the following shall be avoided (1):

(i) An alteration of course to port for a vessel forward of the beam, other than for a vessel being overtaken (1/2);

(ii) An alteration of course toward a vessel abeam or abaft the beam (1/2).

b. You assess the contact above to be drawing rapidly left with a CPA in excess of 2nm. You now hold a small power driven vessel visually at green 40 at 9c; you are 60° on her port bow and her bearing is drawing very slowly left. Does Rule 19 still apply? Explain how you would manoeuvre for this contact. (R. 19a, 15) (3)

This Rule applies to vessels not in sight of one another when navigating in or near an area of restricted visibility (1). Therefore, Rules in part B section II apply (1). This is therefore a crossing situation and you should manoeuvre in accordance with Rule 15. In this case, an alteration of course to starboard or a reduction in speed would be acceptable (1).

c. You make a shipping report to the CO and take the correction action in good time. Continuing along your nav track, you hear a fog signal to port consisting of one prolonged blast followed by two short blasts. What type of vessel could this be? (R. 35c) **(3-half mark each)**

**A vessel not under command,
a vessel restricted in her ability to manoeuvre,
a vessel constrained by her draft,
a sailing vessel,
a vessel engaged in fishing
a vessel engaged in towing or pushing another vessel**

d. You check the radar and hold no steady contacts on the port side other than a faint echo at R20, range 5 cables. You cannot track the echo effectively or generate an accurate CPA using the radar. What action should you take in accordance with the rules? (R. 19e) **(2)**

Except where it has been determined that a risk of collision does not exist(1/2), every vessel which hears apparently forward of her beam the fog signal of another vessel, or which cannot avoid a close-quarters situation with another vessel forward of her beam(1/2), shall reduce her speed to be the minimum at which she can be kept on her course(1/2). She shall if necessary take all her way off and in any event navigate with extreme caution until danger of collision is over(1/2).

The contact closes to 1 mile and you sight the following lights:

R
G

R

e. What type of vessel is this? (R. 25a/c) **(1)**

A sailing vessel underway:

5. The vessel passes clear down your port side and you notice that visibility is rapidly improving as you exit the fog bank. You assess visibility as 10nm and inform the CO who is happy for you now to increase speed to meet the required SOA. As you come up in speed you sight a small cargo vessel at G30, range 3nm. You assess that you are 40 degrees on her port bow and her bearing is steady.

a. In accordance with the Rules what action should you take? (R. 15) **(2)**

When two power-driven(1/2) vessels are crossing so as to involve risk of collision(1/2), the vessel which has the other on her own starboard side (1/2) shall keep out of the way and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel(1/2).

b. What should the cargo vessel do? (R. 17) **(1)**

Where one of two vessels is to keep out of the way, the other shall keep her course and speed.

c. The vessel is now at 5 cables and you have not taken any action to avoid collision. You decide, given the closing range, that urgent action is required to avoid collision and inform the CO. What could you decide to do? (R. 8e) **(1)**

If necessary to avoid collision or allow more time to assess the situation, a vessel may slacken her speed or take all way off by stopping or reversing her means of propulsion.

d. How should any action be taken to avoid collision in accordance with the rules? (R. 8a) **(2)**

Any action to avoid collision shall be taken in accordance with the Rules of this Part(1/2) and shall, if the circumstances of the case admit, be positive(1/2), made in ample time (1/2) and with due regard to the observance of good seamanship(1/2).

6. The cargo vessel takes avoiding action in accordance with Rule 17b and makes a bold alteration of course to starboard. A collision narrowly avoided, the vessel passes 3 cables clear down your stbd side. The CO comes to the bridge to debrief you on the recent incident. He stresses the importance of the Rules and the requirement to act early to generate 'thinking time' in a developing situation. You both reflect on the shipping situation and agree that a simple alteration of course would have resolved the problem early. If taking action to avoid collision by alteration of course alone, what should you consider? **(R. 8c) (2)**

If there is sufficient sea room, alteration of course alone may be the most effective action to avoid a close-quarters situation (1/2) provided that it is made in good time (1/2), is substantial (1/2) and does not result in another close-quarters situation (1/2).

7. You approach the formation of SNMG1 units and note all other ships are in station.

a. What does ATP1G state about how ships not in station should manoeuvre? (ATP1G) **(1)**

Ships that are not in station are not to hamper those in station.

b. What should ships in station consider in such a situation? (ATP1G) **(1)**

Ships in station should not stubbornly maintain their course and speed if danger of collision exists

c. Once in station yourself you note, the Italian aircraft carrier CAVOUR has commenced fixed wing operations. What is the order of priority applicable to Allied naval ships? (ATP1G) **(6)**

Priority is in the following order:

(1) Helicopters.

(2) MCM Units.

(3) Ships Engaged in Replenishment.

(4) Ships Engaged in Launching or Recovery of Landing Craft Utilities.

(5) Ships Engaged in the Launching or Recovery of Aircraft.

(6) Ships Engaged in the Launching or Recovery of Arrays.

d. Do the above priorities take precedence over the International Regulations for Preventing Collisions at Sea? (ATP1G) **(1)**

Yes, the priorities above are applicable to Allied naval ships and overrule the International Regulations for Preventing Collisions at Sea in the circumstances described.

The exercise completes at sunset. As you break away you note that the ITS CAVOUR does not have navigation lights in standard positions.

8. What do the Rules say about the operation of special rules with respect to additional station or signal lights? (R. 1c) **(2)**

Whenever the Government concerned shall have determined that a vessel of special construction or purpose cannot comply fully with the provision of any of these rules with respect to the number, position, range or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signaling appliances(1), without interfering with the special function of the vessel, such vessel shall comply with such other provisions in regard to the number, position, range or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signaling appliances, as her Government shall have determined to be the closest possible compliance with these rules in respect to that vessel (1).

You are preparing night orders after a long day on the bridge when the CAVOUR sends Flag 5 with a time and you hear her sounding 6 short blasts.

9. What does this mean? (ATP 1G) **(1)**

Machinery breakdown

You report this to the CO and head down below for dinner.

You are tasked to proceed alongside to collect another batch of SOLAS stores prior to heading East. After a quiet evening you are up on the bridge early to make final preparations for SSD.

10. You sight a vessel in the early morning twilight proceeding towards the harbour showing white over red masthead lights in addition to sidelights and sternlight.

a. What can you deduce about this vessel? (R. 29a) **(2)**

Pilot vessel on duty, underway.

b. What identity sound signal would this vessel sound in restricted visibility? (R. 35k)
(2)

Four short blasts.

Total / 62